Topic quiz: Projectile motion

IBDP Physics SL · Grade 11 · A.1 Kinematics · 10 multiple-choice questions

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Name: Class: Date:

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use g = 9.81 m s^{-2} and show any working next to the question.
Check yourself online: the same quiz with instant feedback and explanations — https://school.taneski.com/physics/dp/quiz-projectile-motion/

1	2	3	4	5	6	7	8	9	10

Question 1

A ball is dropped from a bench. At the same moment, an identical ball rolls horizontally off the same bench. Air resistance is negligible. Which ball reaches the floor first?

(A) Both reach the floor at the same time
(B) The dropped ball
(C) The ball that rolls off
(D) It depends on how fast the second ball rolls

Question 2

A stone is dropped from rest and falls freely for 2.00 s. How far does it fall?

(A) 9.81 m
(B) 19.6 m
(C) 39.2 m
(D) 4.91 m

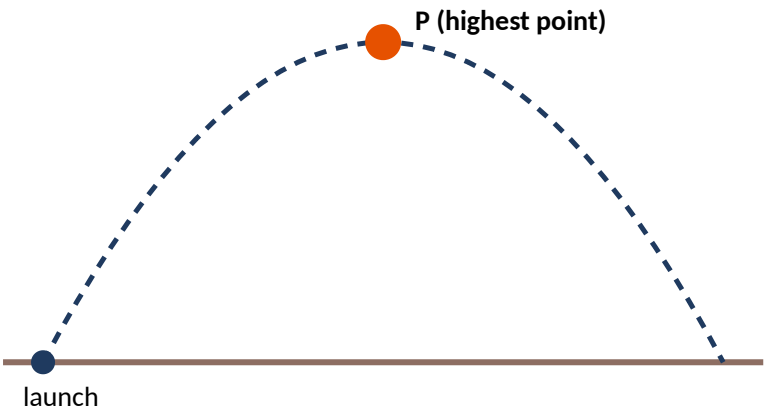
Question 3

A ball rolls off a horizontal bench 0.80 m high with a speed of 2.5 m s^{-1}. How far from the foot of the bench does it land?

(A) 0.64 m
(B) 0.71 m
(C) 1.0 m
(D) 2.0 m

Question 4

A ball is launched at an angle above level ground. Air resistance is negligible. Which statement about the ball at its highest point P is correct?



(A) Its velocity is zero and its acceleration is zero

(B) Its velocity is zero and its acceleration is 9.81 m s^{-2} downwards

(C) Its velocity is horizontal and its acceleration is zero

(D) Its velocity is horizontal and its acceleration is 9.81 m s^{-2} downwards

Question 5

A ball is kicked from level ground at 20.0 m s^{-1} at 30° above the horizontal. What is its maximum height?

(A) 5.10 m

(B) 10.2 m

(C) 15.3 m

(D) 20.4 m

Question 6

At a certain speed, a ball launched from level ground at 30° lands 35 m away. At which other launch angle does the same speed give the same range?

(A) 15°

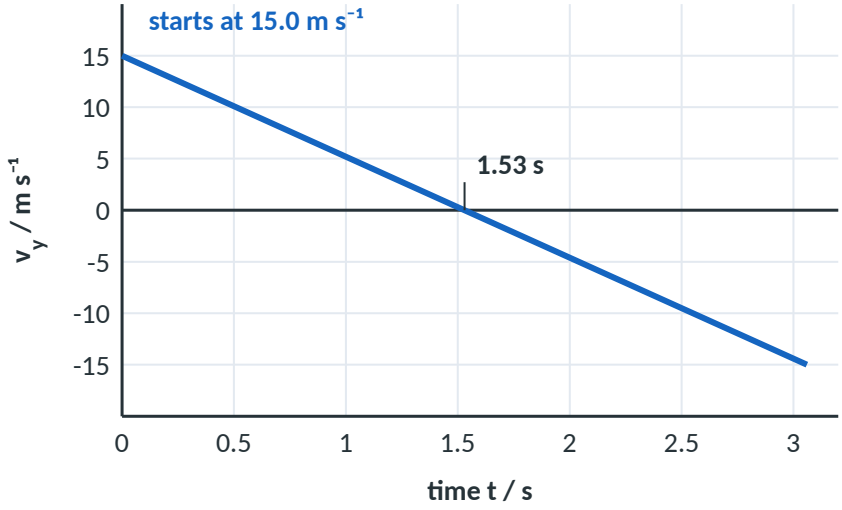
(B) 45°

(C) 60°

(D) 75°

Question 7

The graph shows the vertical velocity v_y of a ball thrown straight upwards (up is positive). What is the maximum height the ball reaches above the launch point?



(A) 0 m

(B) 11.5 m

(C) 15.0 m

(D) 22.9 m

Question 8

A projectile moves without air resistance. What is the change in its velocity between t = 0.5 s and t = 1.5 s?

(A) Zero, because the horizontal velocity is constant

(B) 9.81 m s^{-1} horizontally

(C) It depends on the launch angle

(D) 9.81 m s^{-1} vertically downwards

Question 9

A marble rolls horizontally off a table 1.25 m high. A cup on the floor is 1.50 m from the edge of the table. What launch speed lands the marble in the cup?

(A) 1.20 m s^{-1}

(B) 2.97 m s^{-1}

(C) 4.20 m s^{-1}

(D) 5.94 m s^{-1}

Question 10

Compared with its path in a vacuum, how does air resistance change the path of a thrown tennis ball?

(A) Shorter range and lower maximum height; the ball comes down more steeply than it went up

(B) Same range, but a lower maximum height

(C) Shorter range and lower maximum height; the path is still symmetrical

(D) Longer range, because the air supports the ball

End of quiz · Score: _____ / 10